

LA SERENA, Chile

WMO: 854880

Lat: 29.9175S

Lon: 71.2011W

Elev: 466

StdP: 14.45

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.0	43.4	34.1	29.2	47.8	37.4	33.3	47.2	15.2	56.6	14.0	56.4	5.6	120	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.3	72.8	62.9	71.3	62.3	69.8	61.7	63.9	70.5	63.0	69.3	62.2	68.3	11.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	81.9	66.8	60.3	79.5	66.1	59.3	76.7	65.2	29.5	70.8	28.8	69.7	28.2	68.6	68.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.8	14.3	13.3	DB	38.4	76.2	2.3	2.6	36.7	78.1	35.4	79.6	34.1	81.1	32.5	83.0	(4)
(5)			WB	35.6	65.5	2.2	1.6	34.0	66.6	32.7	67.6	31.4	68.5	29.8	69.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.7	63.9	64.1	61.8	58.2	55.5	53.1	52.2	53.2	54.5	56.3	58.6	61.2	(6)
(7)		DBStd	4.75	2.13	2.23	2.53	2.92	2.90	2.89	2.80	2.53	2.32	2.02	1.85	2.15	(7)
(8)		HDD50	17	0	0	0	0	0	5	9	3	0	0	0	0	(8)
(9)		HDD65	2700	47	39	103	203	293	356	396	366	315	271	193	118	(9)
(10)		CDD50	2826	432	395	367	247	172	99	78	102	135	194	257	348	(10)
(11)		CDD65	36	14	15	5	0	0	0	0	0	0	0	0	2	(11)
(12)		CDH74	22	10	8	2	0	0	1	0	0	0	0	0	1	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.9	6.6	6.3	5.7	5.3	5.2	5.4	5.5	5.8	6.3	6.4	6.7	(14)	
(15)	Precipitation	PrecAvg	3.1	0.0	0.0	0.0	0.0	0.3	0.9	0.9	0.6	0.2	0.1	0.0	0.0	(15)
(16)		PrecMax	8.8	0.1	0.1	0.6	1.0	2.7	6.3	7.0	3.8	1.2	0.8	0.2	0.4	(16)
(17)		PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.4	0.0	0.0	0.1	0.2	0.6	1.3	1.5	0.9	0.3	0.2	0.0	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	75.4	73.2	69.9	68.2	65.7	65.5	66.4	66.3	67.7	69.1	71.9	(19)
(20)			MCWB	64.3	64.9	63.0	60.3	58.5	56.3	55.1	55.7	56.9	56.6	58.9	60.7	(20)
(21)		2%	DB	72.6	72.9	70.6	67.9	65.2	62.8	62.2	62.8	63.6	65.0	67.1	69.8	(21)
(22)			MCWB	63.1	63.4	62.0	59.7	57.4	55.3	54.3	54.9	55.4	56.2	58.0	60.8	(22)
(23)		5%	DB	71.2	71.2	69.2	66.2	62.9	60.8	59.9	60.7	61.9	63.7	65.8	68.1	(23)
(24)			MCWB	62.6	62.7	61.4	58.7	56.3	54.6	53.3	54.0	54.6	55.7	57.6	60.1	(24)
(25)		10%	DB	69.6	69.7	67.6	64.3	61.0	59.0	58.3	59.0	60.4	62.3	64.2	66.6	(25)
(26)			MCWB	61.9	62.1	60.7	58.1	55.6	53.8	52.6	53.2	54.0	55.2	57.1	59.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.9	65.6	64.4	62.1	60.3	58.9	57.1	57.9	58.3	58.6	60.3	62.7	(27)
(28)			MCDWB	72.9	73.3	70.7	67.0	65.1	63.3	62.2	63.1	63.6	64.6	66.0	68.5	(28)
(29)		2%	WB	63.9	64.0	62.9	60.8	58.6	56.8	55.6	56.1	56.6	57.5	59.2	61.5	(29)
(30)			MCDWB	70.5	70.8	68.9	65.7	63.2	60.6	60.1	60.6	61.9	63.2	65.1	67.5	(30)
(31)		5%	WB	63.0	63.0	61.9	59.8	57.4	55.6	54.4	55.0	55.5	56.5	58.3	60.7	(31)
(32)			MCDWB	69.5	69.5	67.5	64.5	61.6	59.4	58.6	59.3	60.4	62.0	63.9	66.6	(32)
(33)		10%	WB	62.1	62.4	61.0	58.8	56.2	54.4	53.3	53.8	54.5	55.7	57.4	59.9	(33)
(34)			MCDWB	68.3	68.4	66.3	63.3	60.3	58.2	57.2	58.1	59.0	60.8	62.8	65.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.3	11.3	11.2	11.4	11.4	12.3	12.9	12.4	12.1	12.1	11.5	11.4	(35)
(36)			MCDBR	13.0	12.9	12.9	13.8	13.9	14.7	15.4	15.6	15.1	15.0	14.0	12.9	(36)
(37)			MCWBR	6.1	6.3	6.7	7.8	8.5	9.7	10.0	10.1	9.0	8.2	7.4	6.4	(37)
(38)		5% WB	MCDBR	12.2	11.8	11.9	12.1	12.9	13.0	14.0	14.0	13.4	13.3	12.3	12.1	(38)
(39)			MCWBR	6.1	6.0	6.4	7.2	8.3	9.0	9.6	9.5	8.4	7.8	6.9	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.379	0.368	0.354	0.342	0.325	0.308	0.307	0.317	0.336	0.333	0.339	0.354	(40)	
(41)		taud	2.399	2.426	2.457	2.478	2.512	2.566	2.555	2.511	2.440	2.452	2.460	2.439	(41)	
(42)		Ebn at Noon	305	303	296	284	274	272	278	288	298	310	315	314	(42)	
(43)		Edh at Noon	40	38	35	31	27	24	25	29	35	37	38	39	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2484	2299	1910	1442	1024	891	974	1233	1630	2069	2379	2533	(44)	
(45)		RadStd	122	96	121	113	92	85	74	101	101	141	124	126	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	+1.28	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page